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Panel Data Econometrics using Stata

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Replication Paper

"IMPACT OF DOUBLE TAXATION TREATIES ON FOREIGN DIRECT INVESTMENT: EVIDENCE FROM LARGE DYADIC PANEL DATA"

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MOTIVATION AND RESEARCH QUESTION

The Double Taxation Treaty is the imposition of comparable taxes in at least two countries on the same taxpayer concerning the same subject matter and for identical periods.
The first version of DTT was published in 1928 by the Group of Experts, which had been convoked by the League of Nations in 1921, and today the original version serves as the foundation of the DTT. (Graetz and O'Hear, 1997). In 2010, the worldwide network of bilateral tax treaties counted more than 18,000 DTTs[1].

In the paper "The impact of DTTs on FDI: Evidence from large dyadic panel data", Barthel, Busse and Neumayer analyze whether the implementation of a DTT leads to a greater amount of Foreign Direct Investment between the two countries at hand.
In the case this holds, the benefits associated to establishing a DTT would outweigh the costs as a result of the economic growth opportunities that arise from Foreign Direct Investment.
It is also important to note that the Double Taxation Treaty on foreign earned income is one of the many obstacles facing cross-border investments considering the negotiation and ratification process ties up administrative resources and may also conflict with domestic tax law.

However, on the other hand, DTTs help to combat tax evasion and avoidance, making information from one contracting state available to the other contract partner, as well as help to reduce costly international tax competition from tax havens.
One of the main points of this paper is that the harmonization of these treaties mitigates the uncertainty an investor faces when dealing with a foreign fiscal system, thus encouraging Foreign Direct Investment.
The purpose of this research is to address some limitations raised from existing studies.
Previous analyses have suffered from the absence of observations on bilateral Foreign Direct Investment, or if they had used bilateral FDI data, they have suffered from a small and unrepresentative dyad year sample.
The authors test the effect of DTTs on FDI in a dyadic country dataset in which both developed and developing countries are broadly represented over a long period of time.

[1] Calculation based on 192 UN members

SAMPLE AND DATA VARIABLES

SAMPLE

The authors implement dyadic[2] fixed effect estimation measuring FDI in stocks, using 30 FDI source countries (countries where the multinational enterprises have their domicile), 10 of which are developing countries, and 105 host countries (countries where the FDI takes place), 84 of which are developing countries.

To achieve a large and representative sample of dyadic FDI, they had to follow several steps. First, they purchased the data from the UNCTAD 2008, mainly bilateral inward FDI stocks information. For those dyads UNCTAD did not report any inward FDI stock data, they reversed the outward FDI stock data (outward FDI stock are highly correlated with the first ones (r=0.86)).

For around half of all dyads in the sample, there were no reported FDI stock data at any point in time. So, for all those dyads where FDI stocks were not reported, they set the FDI stock to zero, since they were fairly confident that there were no bilateral FDI stocks existent.

To allow for an easy interpretation of estimated coefficients as elasticities, they take the natural log of FDI stocks.

Then, the authors have regressed the effect of having a DTT on the logarithm of FDI stock using three different estimations: Static Fixed Effect, Dynamic Fixed Effect, Arellano Bond GMM.

VARIABLES

The dependent variable is *lnfdiinstock* referring to the logarithm of FDI stock.

dtt is the explanatory dummy variable of main interest that shows the presence of a Double Taxation Treaty. They use the year of the effectiveness of the DTT right after the ratification process, as the start of the DTT.

As other control variables, they included:

hostgdptotln which stands for the logarithm of total host GDP in nominal USD to control for the market size, which is expected to have a positive influence on FDI.

hostgdppcln is the logarithm of host GDP per capita in nominal UDS to control for the mean purchasing power of domestic consumers.

hostinflln which is the logarithm of the inflation rate in the host country in percent, measured by the GDP deflator. This has been used in the regression as a proxy for macroeconomic distortions that have a negative impact on FDI.

[2] Related to interaction between two countries

hosttrade is the ratio of a sum of exports and imports to GDP in the host country, to control for the openness of trade. It serves as a proxy for the openness towards foreigners and it shows a positive attitude toward globalization.

rta dummy which stands for a regional trade agreement and takes on the value of 1 if the home and host country have signed a bilateral or a common trade agreement and it has a positive effect on FDI.

bit is a binary variable that controls for the existence of a bilateral investment treaty.

The analysis covers a period from 1978 to 2004, and they added year specific time dummy variables to all estimations to account for any trends in the total FDI.

Descriptive Variable Statistics

Variable	Obs	Mean	Std. Dev.	Min	Max
lnfdiinstock	33,066	1.13045	4.138679	-2.302585	12.53398
dtt	33,066	.3449465	.475358	0	1
dttage	33,066	7.235257	12.67564	0	79
bit	33,066	.154993	.3619035	0	1
bitage	33,066	1.576	5.002215	0	43
hostgdptotln	33,066	24.20884	2.08983	19.63473	30.02448
hostgdppcln	33,066	7.705668	1.567403	4.467087	10.57144
year					
1978	33,066	.004113	.0640015	0	1
1979	33,066	.0058671	.0763728	0	1
1980	33,066	.0149398	.1213139	0	1
1981	33,066	.0169661	.1291461	0	1
1982	33,066	.0154237	.1232326	0	1
1983	33,066	.0143955	.1191161	0	1
1984	33,066	.0152422	.1225168	0	1
1985	33,066	.0136696	.1161171	0	1
1986	33,066	.0169358	.129033	0	1
1987	33,066	.0217746	.1459491	0	1
1988	33,066	.0246477	.1550513	0	1
1989	33,066	.025434	.1574416	0	1
1990	33,066	.0288211	.1673061	0	1
1991	33,066	.0342043	.1817564	0	1
1992	33,066	.039406	.1945619	0	1
1993	33,066	.0422488	.2011594	0	1
1994	33,066	.0463618	.2102706	0	1
1995	33,066	.0505655	.2191121	0	1
1996	33,066	.0541644	.2263452	0	1
1997	33,066	.0549507	.2278875	0	1
1998	33,066	.0578237	.2334134	0	1
1999	33,066	.0627835	.2425769	0	1
2000	33,066	.0696183	.2545066	0	1
2001	33,066	.068076	.25188	0	1
2002	33,066	.0695579	.2544042	0	1
2003	33,066	.06983	.2548642	0	1
2004	33,066	.0621787	.2414834	0	1
hostinflln	33,066	2.4712	1.650454	-4.066673	10.11364
hosttrade	33,066	70.34859	36.29081	10.07903	233.9413
rta	33,066	.1468276	.3539393	0	1

The methods implemented by the authors, where based on the dyadic fixed effect estimator, where each dyad of the countries has its own intercept.

The authors in this paper follow three different specifications:
The first one is the static fixed effect model where no lagged dependent variable is included in the regression.
The second model the dynamic fixed effect model where the lagged dependent variable is included, while the third model is the GMM Arellano Bond estimation.

According to the authors, the static model tends to overestimate the effect of Bilateral Investment Treaties and Double Taxation Treaties as it ignores the dynamic nature of the FDI.
This means that running a static regression to isolate the impact of the DTTs on Foreign Direct Investment would be biased (also because, as evident in the GMM estimation, the lagged dependent variable is statistically significant).
The second regression model, the dynamic fixed effect model which includes a lagged dependent variable as an explanatory variable cannot be used either because, by construction, the lagged dependent variable correlates with the error term, underlying the endogeneity problem.

METHODS

The purpose of this replication paper is to criticize the estimation models chosen by the authors in favour of the GMM Arellano Bond regression, which seems to be the more suitable model to implement.

We are dealing with a dynamic panel model, which represents one of the main causes of endogeneity in the explanatory variables.
One of the most important assumptions for unbiasedness and consistency of OLS estimation in linear panel regression models is the strict exogeneity of explanatory variables. But this assumption is violated if a lagged dependent variable is included in the regression and if there is serial correlation in the error terms.

The purpose of a dynamic linear panel data model is to analyze the dependence of y_{it} on its past values: this is indicated as an endogenous model since this is highly dependent on the past.

The endogeneity is defined as the existence of correlation between the dependent variable and the error term.

The model allows us to distinguish the true state dependence, about how the experience of the event in the past can influence the occurrence of the same event in the future, from indirect effects of the unobserved heterogeneity and from direct effects of the observed heterogeneity in the explanatory variables.

One of most general form of the dynamic linear panel model with unobserved heterogeneity is

$$y_{it} = \gamma_1 y_{i,t-1} + \cdots + \gamma_p y_{i,t-p} + \beta_1 x_{it1} + \cdots + \beta_k x_{itk} + \cdots + \alpha_i + v_{it}$$

This includes the special case with the AR(1) model including additional explanatory variables

$$y_{it} = \gamma_1 y_{i,t-1} + \beta' x_{it} + \alpha_i + v_{it}$$

With $x_{it} = (x_{it1}, \dots, x_{itk})'$ and $\beta = (\beta_1, \dots, \beta_k)'$

In this model we control for the decomposed error component which contains α_i time constant fixed effect and v_{it} time varying idiosyncratic error.

ESTIMATION MODELS

With this model, the conventional OLS estimator is inconsistent because of the $\varepsilon_{it} = \alpha_i + v_{it}$ composite error which is correlated with the $y_{i,t-1}$ in each time period, due to the presence of individual effects.

Since the $cov\left(lnfdiinstock_{i,t-1}, \alpha_i\right) \neq 0$, the solution may be simply to eliminate the fixed effect α_i.

The fixed effects estimator (also called within transformation) results in:

$$y_{it} - \overline{y_i} = \gamma_1\left(y_{i,t-1} - \overline{y_i}\right) + \beta'\left(x_{i,t} - \overline{x_i}\right) + (v_{it} - \overline{v_i})$$

The estimator gets rid of the fixed effects. However, since y_{it} depends on v_{it}, and $y_{i,t-1}$ depends on $v_{i,t-1}$, $y_{i,t-1}$ is correlated with $\overline{v_i}$, because $\overline{v_i}$ is a function of all v_i in every time period.

This is because $\overline{v_i}$ contains $v_{i,t-1}$ which correlates with $y_{i,t-1}$, so, the dependent variable would be endogenous given that $corr\left((lnfdiinstock_{i,t-1} - \overline{lnfdiinstock_i}), (v_{it} - \overline{v_i})\right) \neq 0$

The OLS estimation in time-demeaned equation results in inconsistent estimators.

As the authors pointed out, the fixed effects estimation in the dynamic model leads to the Nickell bias (1981) when "a typical set of panel data has a rather large number of individuals and a rather small number of time periods and it is in this circumstances that the biases are most serious".
This panel data contemplates a number of 30 time periods and a very large number of observations and according to Judson and Owen (1999), using the Monte Carlo Simulation they have proved that T = 30 is considered as a small time period frame.
This bias vanishes only when the number of time periods T of the panel becomes large, thus suggesting the use of another type of estimation technique.

For the random effects estimation, the situation is basically the same.
The quasi-demeaned equation would result in

$$y_{it} - \lambda\overline{y_i} = \gamma\left(y_{i,t-1} - \lambda\overline{y_i}\right) + \beta'\left(x_{i,t} - \lambda\overline{x_i}\right) + (\varepsilon_{it} - \lambda\overline{\varepsilon_i})$$

where $\varepsilon_{it} = \alpha_i + v_{it}$
Since by construction $lnfdiinstock_{i,t-1}$ is correlated with α_i, so it would make no sense to use the random effects estimation method since one regressor is correlated with the fixed effect.
Another estimation model might be in the form of first differenced linear dynamic panel model, which helps to eliminate the fixed effects from the regression.
First differencing results in inconsistent OLS estimates due to the correlation between $\Delta lnfdiinstock_{i,t-1}$ and $\Delta v_{i,t}$.
This is because $\Delta lnfdiinstock_{i,t-1}$ depends on $v_{i,t-1}$, and this leads to the fact that the first differenced explanatory variable $\Delta lnfdiinstock_{i,t-1} = lnfdiinstock_{i,t-1} - lnfdiinstock_{i,t-2}$ is correlated with the error term $\Delta v_{i,t} = v_{i,t} - v_{i,t-1}$
The standard approach is to find a suitable instrument that is strong correlated with the endogenous variable, but uncorrelated with the error terms.
The authors have argued in the paper that they could not find any instrument external to the system for the use of the IV estimation that would fulfil the necessary requirement of being relevant, redundant and fulfilling the identifying restriction.
Given this, the suitable instruments may be internal to the dataset and these should always satisfy the two conditions for valid instruments.
In the model, $\Delta v_{i,t}$ is uncorrelated with $lnfdiinstock_{i,t-k}$ for k higher or equal to 2.
According to the intuitions of Anderson and Hsiao in 1982, to control endogeneity one solution might be to use $\Delta lnfdiinstock_{i,t-2}$ or $lnfdiinstock_{i,t-2}$ as instruments for $\Delta lnfdiinstock_{i,t-1}$

The lagged levels of endogenous variable, three or more period before, can be used as instruments, and if the panel includes three or more periods, we will have more available instruments than unknown parameters.
It is important to note that the Method of Moments cannot be used either, because it is not asymptotically efficient since T is higher than 3 and the model is overidentified.
The method mentioned above is the Generalized Method of Moments which gives the possibility to define more than one moment condition per parameter to be estimated.
Two models can be implemented using GMM: "Arellano Bond" estimator and the "Arellano Bover" estimator. The first one is also known as the "Difference GMM", while the second "System GMM".
The "Arellano Bover" estimator includes lagged differences as instruments for the level equation. It is helpful because it allows for more instruments and can improve efficiency, but it cannot be used since it does not delete the fixed effects, which is a problem as it does not eliminate the endogeneity issue.

With the "Arellano Bond" GMM estimation technique, we can use further lags back as instrument variables for the first lag of the dependent variable, transforming the data to remove the fixed effects.

RESULTS

Before running any regression command, it is important to define in Stata that we are dealing with panel data.

```
. tsset dyadid year
       panel variable:  dyadid (unbalanced)
        time variable:  year, 1977 to 2006
                delta:  1 unit
```

The command `tsset` declares that the data are specifically cross-sectional time-series data, which contain one time series for each value of the panel data model.
As noticeable, this results in an unbalanced panel meaning that some data are missing for some observations for some time periods.

The starting point for the Arellano Bond estimation is the first-differenced dynamic linear panel model.

As the authors pointed out, by first differencing our dummy of interest (*dtt*), the output table will estimate only the effect of DTTs in the first year of becoming effective, whereas we are interested in the effect that a Double Taxation Treaty employs over its entire lifetime.
Indeed, for a dummy variable, this creates the problem that the first differenced variable is zero at first, is one only in the year of becoming effective and zero again in all subsequent years.
In order to avoid this complicacy, they have replaced this dummy variable with *dtt_age*, a variable that measures the years passed since becoming effective, with the year of conclusion set on one.
Since *dtt_age* is explicitly treated as endogenous, the first differenced *dtt_age* and *bit_age* variables are instrumented with their lagged levels.

The model analyzed in the paper takes the form of

$$\begin{aligned} lnfdiinstock_{i,t} = {} & \gamma_1 lnfdiinstock_{i,t-1} + \beta_1 dttage_{i,t} + \beta_2 bitage_{i,t} + \beta_3 hostgdptotln_{i,t} \\ & + \beta_4 hostgdppcln_{i,t} + \beta_5 hostinflln_{i,t} + \beta_6 hosttrade_{i,t} + \beta_7 rta_{i,t} \\ & + \beta_k yeardummy_{i,t} + \alpha_i + v_{i,t} \end{aligned}$$

Under Arellano Bond estimation, the model would result in:

$$\begin{aligned} \Delta lnfdiinstock_{i,t} = {} & \gamma_1 \Delta lnfdiinstock_{i,t-1} + \beta_1 \Delta dttage_{i,t} + \beta_2 \Delta bitage_{i,t} + \beta_3 \Delta hostgdptotln_{i,t} \\ & + \beta_4 \Delta hostgdppcln_{i,t} + \beta_5 \Delta hostinflln_{i,t} + \beta_6 \Delta hosttrade_{i,t} + \beta_7 \Delta rta_{i,t} + \beta_k \Delta yeardummy_{i,t} \\ & + \Delta v_{i,t} \end{aligned}$$

Analyzing the structure of the commands, the authors considered:
One stage GMM Arellano Bond estimator (standard errors are adjusted for clustering on dyadic), with
xtabond which fits a linear dynamic panel-data model where the unobserved panel-level effects are correlated with the lags of the dependent variable, known as the Arellano Bond estimator
maxdep (6) maximum 6 lags of dependent variable for use as instruments
maxlags (6) maximum 6 lags of predetermined and endogenous variables for use as instruments
pre(..) predetermined variables included in the models which are explicitly treated as endogenous.
If they are either predetermined and endogenous, the explanatory variables are correlated with the idiosyncratic error in the same and previous periods, but not in future periods.

The model presents 387 instruments.
One disadvantage with the "Difference GMM" is that it magnifies gaps in unbalanced panels: since there are some values of $y_{i,t}$ which are missing, then both $\Delta y_{i,t}$ and $\Delta y_{i,t-1}$ will be missing in the transformed data.

```
. xi: xtabond  lnfdiinstock  hostgdptotln hostgdppcln  hostinflln hosttrade  rta i.year , robust maxldep(6)
  maxlags(6) pre(dttage bitage, endog)
```

Arellano Bond GMM estimation

```
i.year            _Iyear_1977-2006    (naturally coded; _Iyear_1977 omitted)
note: _Iyear_2004 dropped from div() because of collinearity
note: _Iyear_2005 dropped from div() because of collinearity
note: _Iyear_2006 dropped from div() because of collinearity
note: _Iyear_1979 dropped because of collinearity
note: _Iyear_2005 dropped because of collinearity
note: _Iyear_2006 dropped because of collinearity

Arellano-Bond dynamic panel-data estimation     Number of obs      =     25,714
Group variable: dyadid                          Number of groups   =      2,515
Time variable: year
                                                Obs per group:
                                                              min =           1
                                                              avg =    10.22425
                                                              max =          26

Number of instruments =    387                  Wald chi2(34)      =    2350.69
                                                Prob > chi2        =     0.0000
One-step results
                                   (Std. Err. adjusted for clustering on dyadid)
```

lnfdiinstock	Coef.	Robust Std. Err.	z	P>|z|	[95% Conf.	Interval]
lnfdiinstock						
L1.	.2597008	.0674986	3.85	0.000	.127406	.3919956
dttage	.0920396	.0091116	10.10	0.000	.0741812	.109898
bitage	.0218822	.0089115	2.46	0.014	.0044159	.0393484
hostgdptotln	.1476646	.0267108	5.53	0.000	.0953124	.2000168
hostgdppcln	-.0633029	.0626974	-1.01	0.313	-.1861876	.0595817
hostinflln	-.0038319	.0022299	-1.72	0.086	-.0082023	.0005386
hosttrade	.0003803	.0005007	0.76	0.447	-.0006009	.0013616
rta	.0357398	.0334079	1.07	0.285	-.0297385	.101218
_Iyear_1978	-.0481093	.0198302	-2.43	0.015	-.0869757	-.0092429
_Iyear_1980	.0336206	.0197997	1.70	0.090	-.0051861	.0724273
_Iyear_1981	.0294781	.0283953	1.04	0.299	-.0261756	.0851319
_Iyear_1982	.0150924	.0296001	0.51	0.610	-.0429227	.0731076
Iyear 1983	.0092477	.0316259	0.29	0.770	-.052738	.0712334
_Iyear_1984	-.0211647	.0363811	-0.58	0.561	-.0924705	.050141
_Iyear_1985	.0173297	.0389479	0.44	0.656	-.0590067	.0936661
_Iyear_1986	.0352806	.0441788	0.80	0.425	-.0513084	.1218695
_Iyear_1987	.0491689	.0471445	1.04	0.297	-.0432327	.1415705
_Iyear_1988	.0481109	.0500339	0.96	0.336	-.0499537	.1461756
_Iyear_1989	.060743	.0497561	1.22	0.222	-.0367772	.1582631
_Iyear_1990	.0529791	.0557627	0.95	0.342	-.0563138	.1622719
_Iyear_1991	.0755531	.0611512	1.24	0.217	-.044301	.1954072
_Iyear_1992	.0244815	.0591928	0.41	0.679	-.0915342	.1404972
_Iyear_1993	.0352912	.0560051	0.63	0.529	-.0744769	.1450592
_Iyear_1994	.0761736	.0572948	1.33	0.184	-.0361222	.1884694
_Iyear_1995	.0732582	.062144	1.18	0.238	-.0485417	.1950581
_Iyear_1996	.0494681	.0613121	0.81	0.420	-.0707014	.1696376
_Iyear_1997	.0614822	.0637636	0.96	0.335	-.0634921	.1864566
_Iyear_1998	.0738987	.0656221	1.13	0.260	-.0547183	.2025157
_Iyear_1999	.0624829	.0649896	0.96	0.336	-.0648943	.1898602
_Iyear_2000	.082771	.0681803	1.21	0.225	-.05086	.2164021
_Iyear_2001	.0430862	.0680673	0.63	0.527	-.0903233	.1764956
_Iyear_2002	.064507	.0677886	0.95	0.341	-.0683562	.1973701
_Iyear_2003	.0796276	.0700199	1.14	0.255	-.0576088	.2168641
_Iyear_2004	.0578658	.0744522	0.78	0.437	-.0880578	.2037895
_cons	-2.99529	.6186774	-4.84	0.000	-4.207876	-1.782705

```
Instruments for differenced equation
        GMM-type: L(2/7).lnfdiinstock L(2/6).dttage L(2/6).bitage
        Standard: D.hostgdptotln D.hostgdppcln D.hostinflln D.hosttrade
                  D.rta D._Iyear_1978 D._Iyear_1979 D._Iyear_1980
                  D._Iyear_1981 D._Iyear_1982 D._Iyear_1983 D._Iyear_1984
                  D._Iyear_1985 D._Iyear_1986 D._Iyear_1987 D._Iyear_1988
                  D._Iyear_1989 D._Iyear_1990 D._Iyear_1991 D._Iyear_1992
                  D._Iyear_1993 D._Iyear_1994 D._Iyear_1995 D._Iyear_1996
                  D._Iyear_1997 D._Iyear_1998 D._Iyear_1999 D._Iyear_2000
                  D._Iyear_2001 D._Iyear_2002 D._Iyear_2003
Instruments for level equation
        Standard: _cons
```

Arellano Bond estimation may not be suitable if the errors are serially correlated

If we have serial correlation, then we cannot use $lnfdiinstock_{i,t-2}$ as instrument variable because $corr(v_{i,t}, v_{i,t-2}) \neq 0$ and so $corr(v_{i,t}, lnfdiinstock_{i,t-2}) \neq 0$ which contradicts the exogeneity assumption.

With the command estat abond, Stata reports the Arellano Bond test for serial correlation in the first differenced residual (which is an additional check on the appropriateness of the instrument), where the null hypothesis is that $cov(\Delta v_{it}, \Delta v_{it-k}) = 0$ where $k \geq 2$

By default, since the authors have not indicated any information about the arttest, Stata calculates the Arellano Bond test for first- and second-order autocorrelation in the first differenced v_{it}.

Arellano Bond test for serial correlation

```
. estat abond

Arellano-Bond test for zero autocorrelation in first-differenced errors
```

Order	z	Prob > z
1	-6.1451	0.0000
2	-.67641	0.4988

```
H0: no autocorrelation
```

This model should indicate that the first order autocorrelation is present, but the second order is not. Indeed, Δv_{it} is naturally correlated with Δv_{it-1}, but not with Δv_{it-2}

The outcome shows no significant evidence of serial correlation in the first-differenced errors at order 2, so the null hypothesis cannot be rejected.

This test can be computed only when the heteroskedasticity-robust standard errors have been specified.

For Arellano Bond estimations in linear dynamic panel models, we can use a large number of instruments, such that the Sargan test of overidentifying restrictions might be especially useful. The null implies that all instruments are valid (exogenous) and the model is correctly specified. In other words, that the moment restrictions are equal to zero $E[Z_{i,t}, m] = E[Z_{i,t}, \Delta v_{i,t}] = 0$, where Z is the instrument matrix contains the lagged values used as instruments.

Only for a homoscedastic error term, the Sargan test has an asymptotic chi-squared distribution.

```
. quietly xi: xtabond  lnfdiinstock  hostgdptotln hostgdppcln  hostinflln hosttrade  rta i.year , maxldep(6)
`  maxlags(6) pre(dttage bitage, endog)
```

Sargan Test

```
. estat sargan
Sargan test of overidentifying restrictions
        H0: overidentifying restrictions are valid

        chi2(352)     =  1291.772
        Prob > chi2   =    0.0000
```

This output above presents strong evidence against the null hypothesis that the overidentifying restrictions are valid.

Rejecting this null hypothesis implies that we need to reconsider our model or our instruments. When rejecting the Sargan test, it is recommended to apply some restrictions to the generations of instruments.

For doing that we can reduce the maximum number of lags of the dependent variable maxldep, as well as the maximum number of lags of the explanatory variables.

* Iyear_1977 omitted; Iyear_1979, Iyear_2005, Iyear_2006 dropped because of collinearity; Iyear_2004, Iyear_2005, Iyear_2006 dropped from div() because of collinearity

```
xi: xtabond lnfdiinstock hostgdptotln hostgdppcln hostinflln hosttrade rta i.year,
robust maxldep(2) maxlags(2) pre(dttage bitage, endog)
```

(*)

```
Arellano-Bond dynamic panel-data estimation     Number of obs      =     25,714
Group variable: dyadid                          Number of groups   =      2,515
Time variable: year
                                                Obs per group:
                                                              min =           1
                                                              avg =    10.22425
                                                              max =          26

Number of instruments =    133                  Wald chi2(34)      =    2561.04
                                                Prob > chi2        =     0.0000
One-step results
                                   (Std. Err. adjusted for clustering on dyadid)
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lnfdiinstock	Coef.	Robust Std. Err.	z	P>|z|	[95% Conf.	Interval]
lnfdiinstock						
L1.	.339654	.1397624	2.43	0.015	.0657246	.6135833
dttage	.0876682	.0165235	5.31	0.000	.0552829	.1200536
bitage	.0201065	.0110109	1.83	0.068	-.0014745	.0416875
hostgdptotln	.1298863	.0280792	4.63	0.000	.0748521	.1849206
hostgdppcln	-.0758586	.0600355	-1.26	0.206	-.1935259	.0418088
hostinflln	-.0037657	.0022045	-1.71	0.088	-.0080865	.0005551
hosttrade	.0007469	.0004326	1.73	0.084	-.0001009	.0015947
rta	.0562632	.0282101	1.99	0.046	.0009723	.1115541
_Iyear_1978	-.0471834	.0201625	-2.34	0.019	-.0867011	-.0076657
_Iyear_1980	.0163099	.0221352	0.74	0.461	-.0270743	.0596942
_Iyear_1981	.0108216	.0301954	0.36	0.720	-.0483604	.0700036
_Iyear_1982	-.0027112	.0319591	-0.08	0.932	-.0653498	.0599275
_Iyear_1983	-.0078521	.033196	-0.24	0.813	-.0729151	.0572108
_Iyear_1984	-.0383444	.038939	-0.98	0.325	-.1146634	.0379745
_Iyear_1985	-.0064535	.0401603	-0.16	0.872	-.0851662	.0722592
_Iyear_1986	.0043314	.0437502	0.10	0.921	-.0814175	.0900803
_Iyear_1987	.0062274	.0490157	0.13	0.899	-.0898416	.1022964
_Iyear_1988	-.000208	.0525809	-0.00	0.997	-.1032647	.1028486
_Iyear_1989	.0080283	.0556305	0.14	0.885	-.1010054	.117062
_Iyear_1990	-.0007867	.0604152	-0.01	0.990	-.1191982	.1176249
_Iyear_1991	.0161121	.0656692	0.25	0.806	-.1125972	.1448214
_Iyear_1992	-.0407624	.0667808	-0.61	0.542	-.1716504	.0901257

```
_Iyear_1993   -.0311295   .0626317   -0.50   0.619   -.1538853   .0916263
_Iyear_1994    .0083815   .0643927    0.13   0.896   -.1178259   .1345889
_Iyear_1995   -.0010856   .0691172   -0.02   0.987   -.1365528   .1343817
_Iyear_1996   -.0283279   .0732847   -0.39   0.699   -.1719632   .1153074
_Iyear_1997   -.0202294   .0747454   -0.27   0.787   -.1667276   .1262689
_Iyear_1998    -.014074   .0774365   -0.18   0.856   -.1658468   .1376987
_Iyear_1999     -.02421   .0817509   -0.30   0.767   -.1844388   .1360189
_Iyear_2000   -.0074188   .0841818   -0.09   0.930   -.1724121   .1575745
_Iyear_2001   -.0511565   .0829676   -0.62   0.538     -.21377   .1114569
_Iyear_2002   -.0284193   .0828916   -0.34   0.732   -.1908838   .1340452
_Iyear_2003   -.0156629   .0843529   -0.19   0.853   -.1809916   .1496657
_Iyear_2004   -.0406188   .0930174   -0.44   0.662   -.2229295   .1416919
      _cons   -2.476611   .6477458   -3.82   0.000    -3.74617   -1.207053
------------------------------------------------------------------------------
Instruments for differenced equation
        GMM-type: L(2/3).lnfdiinstock L(2/2).dttage L(2/2).bitage
        Standard: D.hostgdptotln D.hostgdppcln D.hostinflln D.hosttrade
                  D.rta D._Iyear_1978 D._Iyear_1979 D._Iyear_1980
                  D._Iyear_1981 D._Iyear_1982 D._Iyear_1983 D._Iyear_1984
                  D._Iyear_1985 D._Iyear_1986 D._Iyear_1987 D._Iyear_1988
                  D._Iyear_1989 D._Iyear_1990 D._Iyear_1991 D._Iyear_1992
                  D._Iyear_1993 D._Iyear_1994 D._Iyear_1995 D._Iyear_1996
                  D._Iyear_1997 D._Iyear_1998 D._Iyear_1999 D._Iyear_2000
                  D._Iyear_2001 D._Iyear_2002 D._Iyear_2003
Instruments for level equation
        Standard: _cons
```

```
. estat abond

Arellano-Bond test for zero autocorrelation in first-differenced errors
```

Order	z	Prob > z
1	-4.2168	0.0000
2	-.58196	0.5606

```
H0: no autocorrelation
```

```
. quietly xi: xtabond lnfdiinstock hostgdptotln hostgdppcln hostinflln hosttrade rta i.year,
  maxldep(2) maxlags(2) pre(dttage bitage, endog)
```

```
. estat sargan
Sargan test of overidentifying restrictions
        H0: overidentifying restrictions are valid

        chi2(98)     =  488.7935
        Prob > chi2  =    0.0000
```

Even restricting the number of lags of the dependent variable and the lags of the predetermined variables, H0 is still rejected at all significant levels, showing that the model is still not valid.

It is possible to argue that for the variable $hostgdptotln_{i,t}$ the z statistic is high, and this may be an indicator of another endogeneity problem. This may be the reason why restricting the number of lagged variables used as instruments does not help to not reject the null hypothesis in the Sargan Test.

Explicitly treating the variable $hostgdptotln_{i,t}$ as endogenous, using its lags as instrument variables will not resolve the problem either.

```
. quietly xi: xtabond lnfdiinstock hostgdptotln hostgdppcln hostinflln hosttrade rta
> i.year, maxldep(2) maxlags(2) pre(dttage bitage, endog) endogenous(hostgdptotln)

. estat sargan
Sargan test of overidentifying restrictions
        H0: overidentifying restrictions are valid

        chi2(123)     =  557.9835
        Prob > chi2   =    0.0000
```

CONCLUSION

In the paper, the authors have examined one important policy instrument, namely, the impact of DTTs on FDI stocks in host economies. The main advantage of this empirical analysis is that they used an impressive number of host and source countries in order to avoid sample selection bias.

The static model is not taken into consideration because it is very likely that the Foreign Direct Investment stocks are highly correlated with the events in the past, and this is also showed in the GMM Arellano Bond estimation where the variable $lnfdiinstock_{i,t-1}$ is statistically significant.

Additionally, when using the GMM Arellano Bond estimator we have to rely on a large dataset, because it is consistent and asymptotically efficient when the sample size tends to infinity. Furthermore, this estimation is valid only if there is no autocorrelation in the idiosyncratic error.

The explanatory variable of interest, *dtt_age* is highly significant in the regression, exhibiting the positive impact of the DTTs on the FDI.

According to the analysis conducted in this replication paper, all three methods implemented by the authors present some issues. As a result, the most suitable one is the GMM Arellano Bond estimator assuming the appropriate modifications have been added in the regression model.
We attempt to address each of the potential issues. First, there is a possibility we are dealing with invalid instruments. We attempt to run the regression with maximum 2 lags of variables instead of maximum 6, in order to correctly specify the model, and we also try to treat the variable $hostgdptotln_{i,t}$ as explicitly endogenous. However, the outcome is still problematic since the null hypothesis of the Sargan Test is rejected.

This indicates that instruments are not exogenous or that the model is not well specified.

However, given IV exogeneity is a common issue in econometrics, we will go forward in supporting the legitimacy of the results given the previous tests we have run, coupled with the explanatory variable of interest being significant.